

Modules (Lmod)

Lmod is a module management system that allows you to dynamically load and unload software environments on Unix-like systems

Follow the steps to install the module.

Step-1: Install Lmod

```
# sudo dnf install Lmod
```

```
[siddharrth@master ~]$ sudo dnf install Lmod
Last metadata expiration check: 2:47:00 ago on Tue 07 Apr 2026 10:42:23 PM EDT.
Dependencies resolved.
=====
Package                                Architecture      Version            Repository          Size
=====
Installing:
Lmod                                   x86_64            8.7.65-3.el8      epel                 294 k
Installing dependencies:
luafilesystem                         x86_64            1.6.3-7.el8       powertools           34 k
lua-posix                             x86_64            33.3.1-9.el8      powertools           175 k
lua-term                              x86_64            0.07-9.el8        epel                 16 k
tcl                                   x86_64            1:8.6.8-2.el8     baseos               1.1 M
Transaction Summary
=====
Install 5 Packages

Total download size: 1.6 M
Installed size: 6.9 M
Is this ok [y/N]: y
```

Step-2: Enable modules

```
# source /etc/profile.d/modules.sh
```

```
# module --version --> Check the module version
```

```
[siddharrth@master ~]$ source /etc/profile.d/modules.sh
[siddharrth@master ~]$ module --version
Lmod Warning: Unknown Option: "e"

Lmod Warning: Unknown Option: "o"

Lmod Warning: Unknown Option: "n"

Modules based on Lua: Version 8.7.65 2025-08-05 10:24 -06:00
by Robert McLay mclay@tacc.utexas.edu
```

Step-3: Verify the module

```
# module avail
```

```
[siddharrth@master ~]$ module avail
----- /usr/share/lmod/lmod/modulefiles/Core -----
lmod    settarg

If the avail list is too long consider trying:

"module --default avail" or "ml -d av" to just list the default modules.
"module overview" or "ml ov" to display the number of modules for each name.

Use "module spider" to find all possible modules and extensions.
Use "module keyword key1 key2 ..." to search for all possible modules matching any of the "keys".
```

Step-4: Install dependencies

```
# sudo dnf groupinstall "Development Tools" -y
```

(installs all essential build tools)

```
[siddharth@master ~]$ sudo dnf groupinstall "Development Tools" -y
[sudo] password for siddharth:
Last metadata expiration check: 2:42:05 ago on Tue 07 Apr 2026 10:42:23 PM EDT.
Dependencies resolved.
=====
Package                                Architecture Version                                Repository                                Size
=====
Installing group/module packages:
asciidoc                               noarch      8.6.10-0.5.20180627gitf7c2274.el8      appstream                                215 k
autoconf                               noarch      2.69-29.el8_10.1                       appstream                                710 k
automake                               noarch      1.16.1-8.el8                           appstream                                713 k
bison                                  x86_64      3.0.4-10.el8                           appstream                                687 k
byacc                                  x86_64      1.9.20170709-4.el8                     appstream                                90 k
ctags                                  x86_64      5.8-23.el8                             appstream                                169 k
diffstat                              x86_64      1.61-7.el8                             appstream                                43 k
elfutils-libelf-devel                 x86_64      0.190-2.el8                            baseos                                    61 k
flex                                   x86_64      2.6.1-9.el8                            appstream                                318 k
gcc                                    x86_64      8.5.0-28.el8_10                        appstream                                23 M
gcc-c++                               x86_64      8.5.0-28.el8_10                        appstream                                12 M
gdb                                    x86_64      8.2-20.el8_0.1                         appstream                                298 k
git                                    x86_64      2.43.7-1.el8_10                        appstream                                92 k
glibc-devel                           x86_64      2.28-251.el8_10.31                    baseos                                    90 k
intltool                              noarch      0.51.0-11.el8                         appstream                                65 k
jna                                    x86_64      4.5.1-5.el8                            appstream                                241 k
libtool                                x86_64      2.4.6-25.el8                           appstream                                708 k
ltrace                                 x86_64      0.7.91-28.el8                          appstream                                159 k
patchutils                             x86_64      0.3.4-10.el8                           appstream                                114 k
perl-Fedora-VSP                        noarch      0.001-9.el8                            appstream                                23 k
perl-generators                       noarch      1.10-9.el8                             appstream                                17 k
=====
```

```
# sudo dnf install openssl-devel bzip2-devel libffi-devel
```

(required to compile Python from source)

```
[siddharth@master 3.12]$ sudo dnf install openssl-devel bzip2-devel libffi-devel
[sudo] password for siddharth:
Last metadata expiration check: 3:16:05 ago on Tue 07 Apr 2026 10:42:23 PM EDT.
Package openssl-devel-1:1.1.1k-15.el8_10.x86_64 is already installed.
Dependencies resolved.
=====
Package                                Architecture Version                                Repository                                Size
=====
Installing:
bzip2-devel                             x86_64      1.0.6-28.el8_10                        baseos                                    224 k
libffi-devel                             x86_64      3.1-24.el8                             baseos                                    28 k
=====
Transaction Summary
=====
Install 2 Packages

Total download size: 252 k
Installed size: 409 k
Is this ok [y/N]: y
Downloading Packages:
(1/2): libffi-devel-3.1-24.el8.x86_64.rpm 27 kB/s | 28 kB 00:01
(2/2): bzip2-devel-1.0.6-28.el8_10.x86_64.rpm 151 kB/s | 224 kB 00:01
-----
Total 116 kB/s | 252 kB 00:02
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing :
  Installing : libffi-devel-3.1-24.el8.x86_64 1/1
  Running scriptlet: libffi-devel-3.1-24.el8.x86_64 1/2
  Installing : bzip2-devel-1.0.6-28.el8_10.x86_64 2/2
  Running scriptlet: bzip2-devel-1.0.6-28.el8_10.x86_64 2/2
  Verifying : bzip2-devel-1.0.6-28.el8_10.x86_64 1/2
  Verifying : libffi-devel-3.1-24.el8.x86_64 2/2

Installed:
bzip2-devel-1.0.6-28.el8_10.x86_64 libffi-devel-3.1-24.el8.x86_64

Complete!
```

Step-5: Install Python manually

```
# sudo mkdir -p /home/siddharth/package/python/3.12
```

```
# cd /home/siddharth/package/python/3.12
```

Step-6: Download Python

```
# wget https://www.python.org/ftp/python/3.12.13/Python-3.12.13.tgz

# tar -xvf Python-3.12.13.tgz

# cd Python-3.12.
```

```
/home/siddharrth/package/python/3.12
[siddharrth@master 3.12]$ ll
total 26628
drwxr-xr-x. 18 siddharrth siddharrth    4096 Mar  3 07:39 Python-3.12.13
-rw-rw-r--.  1 siddharrth siddharrth 27262093 Mar  3 08:16 Python-3.12.13.tgz
[siddharrth@master 3.12]$ cd Python-3.12.13/
[siddharrth@master Python-3.12.13]$ |
```

Step-7: Build and install

```
# ./configure --prefix=/home/siddharrth/package/python/3.12
```

```
[siddharrth@master Python-3.12.13]$ ./configure --prefix=/home/siddharrth/package/python/3.12
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for Python interpreter freezing... ./_bootstrap_python
checking for python3.12... no
checking for python3.12... no
checking for python3.11... python3.11
checking Python for regen version... Python 3.11.13
checking for pkg-config... /usr/bin/pkg-config
checking pkg-config is at least version 0.9.0... yes
checking for --enable-universalsdk... no
checking for --with-universal-archs... no
checking MACHDEP... "linux"
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether gcc accepts -g... yes
checking for gcc option to enable C11 features... none needed
checking how to run the C preprocessor... gcc -E
checking for grep that handles long lines and -e... /usr/bin/grep
checking for stdlib extension module _testmultiphase... yes
checking for stdlib extension module xxsubtype... yes
checking for stdlib extension module _xxtestfuzz... yes
checking for stdlib extension module _ctypes_test... yes
checking for stdlib extension module xxlimited... yes
checking for stdlib extension module xxlimited_35... yes
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/Setup.bootstrap
config.status: creating Modules/Setup.stdlib
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
configure: creating Modules/Setup.local
configure: creating Makefile
```

```
# make -j4
```

```
[siddharrth@master Python-3.12.13]$ make -j4
gcc -pthread -c -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O3 -Wall -std=c11 -Wextra -Wno-unused-parameter -Wno-missing-field-initializers -Wstrict-prototypes -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I. -I./Include -DPy_BUILD_CORE -o Programs/python.o ./Programs/python.c
gcc -pthread -c -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O3 -Wall -std=c11 -Wextra -Wno-unused-parameter -Wno-missing-field-initializers -Wstrict-prototypes -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I. -I./Include -DPy_BUILD_CORE -o Parser/token.o Parser/token.c
gcc -pthread -c -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O3 -Wall -std=c11 -Wextra -Wno-unused-parameter -Wno-missing-field-initializers -Wstrict-prototypes -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I. -I./Include -DPy_BUILD_CORE -o Parser/pegen.o Parser/pegen.c
gcc -pthread -c -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O3 -Wall -std=c11 -Wextra -Wno-unused-parameter -Wno-missing-field-initializers -Wstrict-prototypes -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I. -I./Include -DPy_BUILD_CORE -o Parser/action_helpers.o Parser/action_helpers.c
gcc -pthread -c -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O3 -Wall -std=c11 -Wextra -Wno-unused-parameter -Wno-missing-field-initializers -Wstrict-prototypes -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I. -I./Include -DPy_BUILD_CORE -o Parser/pegen_errors.o Parser/pegen_errors.c
gcc -pthread -c -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O3 -Wall -std=c11 -Wextra -Wno-unused-parameter -Wno-missing-field-initializers -Wstrict-prototypes -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I. -I./Include -DPy_BUILD_CORE -o Parser/parser.o Parser/parser.c
gcc -pthread -c -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O3 -Wall -std=c11 -Wextra -Wno-unused-parameter -Wno-missing-field-initializers -Wstrict-prototypes -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I. -I./Include -DPy_BUILD_CORE -o Parser/string_parser.o Parser/string_parser.c
gcc -pthread -c -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O3 -Wall -std=c11 -Wextra -Wno-unused-parameter -Wno-missing-field-initializers -Wstrict-prototypes -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I. -I./Include -DPy_BUILD_CORE -o Parser/string_parser.o Parser/string_parser.c
```

```
Read.o Python/traceback.o Python/tracemalloc.o Python/getopt.o Python/pystrip.o Python/pystrip.o Python/pystrip.o Python/dtoa.o Python/formatter_unicode.o Python/fileutils.o Python/suggestions.o Python/perf_trampoline.o Python/dynload_shlib.o Modules/config.o Modules/main.o Modules/gcmodule.o Modules/atexitmodule.o Modules/faulthandler.o Modules/posixmodule.o Modules/signalmodule.o Modules/_tracemalloc.o Modules/_codecsmodule.o Modules/_collectionsmodule.o Modules/_errnomodule.o Modules/_io/_iomodule.o Modules/_io/_iobase.o Modules/_io/_fileio.o Modules/_io/_bytesio.o Modules/_io/_bufferedio.o Modules/_io/_textio.o Modules/_io/stringio.o Modules/itertoolsmodule.o Modules/_sre/sre.o Modules/_threadmodule.o Modules/_timemodule.o Modules/_typingmodule.o Modules/_weakref.o Modules/_abc.o Modules/_functoolsmodule.o Modules/_localemodule.o Modules/_operator.o Modules/_stat.o Modules/_symtablemodule.o Modules/_pwdmodule.o Python/deepfreeze/deepfreeze.o Modules/getpath.o Python/frozen.o _lpthread _ldl _lutil -lm
./python -E -S -m sysconfig --generate-posix-vars ; \
if test $? -ne 0 ; then \
    echo "generate-posix-vars failed" ; \
    rm -f ./pybuilddir.txt ; \
    exit 1 ; \
fi
./python -E -c 'import sys ; from sysconfig import get_platform ; print("%s-%d.%d" % (get_platform(), *sys.version_info[:2]))' >platform
The following modules are *disabled* in configure script:
_sqlite3

The necessary bits to build these optional modules were not found:
_curses          _curses_panel    _dbm
_gdbm            _tkinter         _uuid
nis              readline

To find the necessary bits, look in configure.ac and config.log.

Checked 111 modules (31 built-in, 70 shared, 1 n/a on linux-x86_64, 1 disabled, 8 missing, 0 failed on import)
```

sudo make install

```
[siddharrth@master Python-3.12.13]$ sudo make install
[sudo] password for siddharrth:
Creating directory /home/siddharrth/package/python/3.12/bin
Creating directory /home/siddharrth/package/python/3.12/lib
if test "no-framework" = "no-framework" ; then \
    /usr/bin/install -c python /home/siddharrth/package/python/3.12/bin/python3.12; \
else \
    /usr/bin/install -c -s Mac/pythonw /home/siddharrth/package/python/3.12/bin/python3.12; \
fi
if test "3.12" != "3.12"; then \
    if test -f /home/siddharrth/package/python/3.12/bin/python3.12 -o -h /home/siddharrth/package/python/3.12/bin/python3.12; \
    then rm -f /home/siddharrth/package/python/3.12/bin/python3.12; \
    fi; \
    (cd /home/siddharrth/package/python/3.12/bin; ln python3.12 python3.12); \
fi
if test "x" != "x" ; then \
    rm -f /home/siddharrth/package/python/3.12/bin/python3.12-32; \
    lipo \
        -output /home/siddharrth/package/python/3.12/bin/python3.12-32 \
        /home/siddharrth/package/python/3.12/bin/python3.12; \
fi
if test "x" != "x" ; then \
    rm -f /home/siddharrth/package/python/3.12/bin/python3.12-intel64; \
    lipo \
        -output /home/siddharrth/package/python/3.12/bin/python3.12-intel64 \
        /home/siddharrth/package/python/3.12/bin/python3.12; \
fi
# Install macOS debug information (if available)
```

```
To find the necessary bits, look in configure.ac and config.log.
```

```
Checked 111 modules (31 built-in, 70 shared, 1 n/a on linux-x86_64, 1 disabled, 8 missing, 0 failed on import)
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/asyncio
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/collections
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/concurrent
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/concurrent/futures
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/ctypes
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/ctypes/macholib
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/curses
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/dbm
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/email
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/email/mime
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/encodings
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/ensurepip
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/ensurepip/_bundled
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/html
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/http
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/idlelib
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/idlelib/Icons
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/importlib
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/importlib/resources
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/importlib/metadata
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/json
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/lib2to3
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/lib2to3/fixes
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/lib2to3/pgen2
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/logging
Creating directory /home/siddharrth/package/python/3.12/lib/python3.12/multiprocessing
```

```
rm -f /home/siddharrth/package/python/3.12/bin/pydoc3
(cd /home/siddharrth/package/python/3.12/bin; ln -s pydoc3.12 pydoc3)
rm -f /home/siddharrth/package/python/3.12/bin/2to3
(cd /home/siddharrth/package/python/3.12/bin; ln -s 2to3-3.12 2to3)
if test "x" != "x" ; then \
    rm -f /home/siddharrth/package/python/3.12/bin/python3-32; \
    (cd /home/siddharrth/package/python/3.12/bin; ln -s python3.12-32 python3-32) \
fi
if test "x" != "x" ; then \
    rm -f /home/siddharrth/package/python/3.12/bin/python3-intel64; \
    (cd /home/siddharrth/package/python/3.12/bin; ln -s python3.12-intel64 python3-intel64) \
fi
rm -f /home/siddharrth/package/python/3.12/share/man/man1/python3.1
(cd /home/siddharrth/package/python/3.12/share/man/man1; ln -s python3.12.1 python3.1)
if test "xupgrade" != "xno" ; then \
    case upgrade in \
        upgrade) ensurepip="--upgrade" ;; \
        install|*) ensurepip="" ;; \
    esac; \
    ./python -E -m ensurepip \
        $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmp2r2inubm3
Processing /tmp/tmp2r2inubm3/pip-25.0.1-py3-none-any.whl
Installing collected packages: pip
WARNING: The scripts pip3 and pip3.12 are installed in '/home/siddharrth/package/python/3.12/bin' which is not on PATH
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-25.0.1
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager, possibly rendering your system unusable. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv. Use the --root-user-action option if you know what you are doing and want to suppress this warning.
```

- you will come across lots of command lines.

Step-8: Create Module Directory

```
# sudo mkdir -p modules/python
```

Step-9: Create Lua Modulefile

```
# sudo vi modules/python/3.12.lua
```

Replace with this content

“

```
help([[
```

```
Python 3.12 Module
```

```
]])
```

```
whatis("Description: Python 3.12")
```

```
prepend_path("PATH", "/home/siddharrth/package/python/3.12/bin")
setenv("PYTHON_HOME", "/home/siddharrth/package/python/3.12/bin")
“
```

```
help([[
Python 3.12 Module
]])

whatis("Description: Python 3.12")

prepend_path("PATH", "/home/siddharrth/package/python/3.12/bin")
setenv("PYTHON_HOME", "/home/siddharrth/package/python/3.12/bin")
~
```

Step-10: Add Module Path

```
# module use modules/
```

To check – # echo \$MODULEPATH

```
[siddharrth@master ~]$ module use modules/
[siddharrth@master ~]$ echo $MODULEPATH
/home/siddharrth/modules:/etc/modulefiles:/usr/share/modulefiles:/usr/share/modulefiles/Linux:/usr/share/modulefiles/Core:/usr/share/lmod/lmod/modulefiles/Core
```

Step-11: verify the module list

```
# module avail
```

```
[siddharrth@master ~]$ module avail

----- /home/siddharrth/modules -----
python/3.12

----- /usr/share/lmod/lmod/modulefiles/Core -----
lmod      settarg

If the avail list is too long consider trying:

"module --default avail" or "ml -d av" to just list the default modules.
"module overview" or "ml ov" to display the number of modules for each name.

Use "module spider" to find all possible modules and extensions.
Use "module keyword key1 key2 ..." to search for all possible modules matching any of the "keys".
```

Step-12: To load the module

```
# module load python/3.12
```

Check python version - # python3.12

```
[siddharrth@master ~]$ module load python/3.12
[siddharrth@master ~]$ python3.12
Python 3.12.13 (main, Apr 8 2026, 02:13:12) [GCC 8.5.0 20210514 (Red Hat 8.5.0-28)] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> exit
Use exit() or Ctrl-D (i.e. EOF) to exit
>>> exit()
```

Step-13: Try adding another python version and swap to other version (follow steps 4, 6,7,9,11,12)


```
[siddharrth@master ~]$ sudo mkdir -p package/python/3.14 ← Create a new dir. for new version
[sudo] password for siddharrth:
[siddharrth@master ~]$ cd package/python/3.14/ ← Open the dir. to install the package
```

```
[siddharrth@master 3.14]$ sudo wget https://www.python.org/ftp/python/3.14.4/Python-3.14.4.tgz ← Download python 3.14
--2026-04-08 03:28:54-- https://www.python.org/ftp/python/3.14.4/Python-3.14.4.tgz
Resolving www.python.org (www.python.org)... 151.101.192.223, 151.101.128.223, 151.101.64.223, ...
Connecting to www.python.org (www.python.org)|151.101.192.223|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 31148035 (30M) [application/octet-stream]
Saving to: 'Python-3.14.4.tgz.1'
```

```
Python-3.14.4.tgz.1 100%[=====] 29.71M 11.1MB/s in 2.7s
```

```
2026-04-08 03:28:57 (11.1 MB/s) - 'Python-3.14.4.tgz.1' saved [31148035/31148035]
```

```
[siddharrth@master 3.14]$ sudo tar -xvf Python-3.14.4.tgz ← Extract the tar file
```

```
[sudo] password for siddharrth:
Python-3.14.4/
Python-3.14.4/.coveragerc
Python-3.14.4/.devcontainer/
Python-3.14.4/.devcontainer/devcontainer.json
Python-3.14.4/.editorconfig
Python-3.14.4/.mailmap
Python-3.14.4/.pre-commit-config.yaml
Python-3.14.4/.readthedocs.yml
Python-3.14.4/.ruff.toml
Python-3.14.4/Android/
Python-3.14.4/Android/README.md
Python-3.14.4/Android/android-env.sh
Python-3.14.4/Android/android.py
Python-3.14.4/Android/testbed/
Python-3.14.4/Android/testbed/.gitignore
Python-3.14.4/Android/testbed/.idea/
Python-3.14.4/Android/testbed/.idea/inspectionProfiles/
Python-3.14.4/Android/testbed/.idea/inspectionProfiles/Project_Default.xml
Python-3.14.4/Android/testbed/app/
Python-3.14.4/Android/testbed/app/.gitignore
Python-3.14.4/Android/testbed/app/build.gradle.kts
Python-3.14.4/Android/testbed/app/src/
Python-3.14.4/Android/testbed/app/src/androidTest/
Python-3.14.4/Android/testbed/app/src/androidTest/java/
```

```
[siddharrth@master Python-3.14.4]$ sudo ./configure --prefix=/home/siddharrth/package/python/3.14 ← Build
```

```
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for Python interpreter freezing... ./_bootstrap_python
checking for python3.14... no
checking for python3.15... no
checking for python3.14... no
checking for python3.13... no
checking for python3.12... no
checking for python3.11... python3.11
checking Python for regen version... Python 3.11.13
checking for pkg-config... /bin/pkg-config
checking pkg-config is at least version 0.9.0... yes
checking MACHDEP... "linux"
```

```
[siddharrth@master Python-3.14.4]$ sudo make -j4
gcc -pthread -c -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O3 -Wall -std=c11 -Wextra -Wno-unused-parameter -Wno-missing-field-initializers -Wstrict-prototypes -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I./Include/internal/mimalloc -I. -I./Include -DPy_BUILD_CORE -o Programs/python.o ./Programs/python.c
gcc -pthread -c -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O3 -Wall -std=c11 -Wextra -Wno-unused-parameter -Wno-missing-field-initializers -Wstrict-prototypes -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I./Include/internal/mimalloc -I. -I./Include -DPy_BUILD_CORE -o Parser/token.o Parser/token.c
gcc -pthread -c -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O3 -Wall -std=c11 -Wextra -Wno-unused-parameter -Wno-missing-field-initializers -Wstrict-prototypes -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/i
```

```
[siddharrth@master Python-3.14.4]$ sudo make install
Creating directory /home/siddharrth/package/python/3.14/bin
Creating directory /home/siddharrth/package/python/3.14/lib
if test "no-framework" = "no-framework" ; then \
    /bin/install -c python /home/siddharrth/package/python/3.14/bin/python3.14; \
else \
    /bin/install -c -s Mac/pythonw /home/siddharrth/package/python/3.14/bin/python3.14; \
fi
if test "3.14" != "3.14"; then \
    if test -f /home/siddharrth/package/python/3.14/bin/python3.14 -o -h /home/siddharrth/package/python/3.14/bin/pytho
thon3.14; \
then rm -f /home/siddharrth/package/python/3.14/bin/python3.14; \
fi; \
fi
```

```

[siddharrth@master ~]$ sudo vi modules/python/3.14.lua ← Create a modulefile
[siddharrth@master ~]$ cat modules/python/3.14.lua
help[[

Python 3.14 Module

]])

whatis("Description: Python 3.14")

prepend_path("PATH", "/home/siddharrth/package/python/3.14/bin")

setenv("PYTHON_HOME", "/home/siddharrth/package/python/3.14/")

[siddharrth@master ~]$ module avail ← Check the module list

----- /home/siddharrth/modules -----
python/3.12 (L)  python/3.14 (D)
----- /usr/share/lmod/lmod/modulefiles/Core -----

lmod  settarg

Where:
L:  Module is loaded
D:  Default Module

If the avail list is too long consider trying:

"module --default avail" or "ml -d av" to just list the default modules.
"module overview" or "ml ov" to display the number of modules for each name.

Use "module spider" to find all possible modules and extensions.
Use "module keyword key1 key2 ..." to search for all possible modules matching any of the "keys".

[siddharrth@master ~]$ module list ← Currently loaded module

Currently Loaded Modules:
  1) python/3.12 ← Previously loaded version

[siddharrth@master ~]$ module load python/3.14 ← To load another version

The following have been reloaded with a version change:
  1) python/3.12 => python/3.14 ← Version Changing

[siddharrth@master ~]$ python3
Python 3.14.4 (main, Apr  8 2026, 03:39:12) [GCC 8.5.0 20210514 (Red Hat 8.5.0-28)] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> exit

```

- Swapped to another version

Step-14: Make it permanent

```

# echo "module use /home/siddharrth/modules" >> ~/.bashrc

# source ~/.bashrc

```



```
[siddharrth@master ~]$ echo "module use /home/siddharrth/modules" >> ~/.bashrc
```

```
[siddharrth@master ~]$ source ~/.bashrc
```

```
[siddharrth@master ~]$ cat .bashrc
```

```
# .bashrc
```

```
# Source global definitions
```

```
if [ -f /etc/bashrc ]; then
```

```
    . /etc/bashrc
```

```
fi
```

```
# User specific environment
```

```
if ! [[ "$PATH" =~ "$HOME/.local/bin:$HOME/bin:" ]]
```

```
then
```

```
    PATH="$HOME/.local/bin:$HOME/bin:$PATH"
```

```
fi
```

```
export PATH
```

```
# Uncomment the following line if you don't like systemctl's auto-paging feature:
```

```
# export SYSTEMD_PAGER=
```

```
# User specific aliases and functions
```

```
module use /home/siddharrth/modules
```

Add in .bashrc to make it permanent

Added here